YIFAN SUN

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EDUCATION

University of Illinois Urbana-Champaign — *Sep 2024 – Present*
PhD in Computer Science, Advisor: Prof. Huan Zhang — *GPA: 4.00 / 4.00*

Columbia University — *Sep 2022 – Feb 2024*
Master in Industrial Engineering and Operations Research (Dual Degree) — *GPA: 4.00 / 4.00*

Tsinghua University — *Sep 2021 – Jun 2024*
Master in Management Information Systems (Dual Degree) — *GPA: 4.00 / 4.00*

Fudan University — *Sep 2017 – Jun 2021*
Bachelor in Statistics — *GPA: 3.94 / 4.00*

RESEARCH INTERESTS

Data-centric Problems in LLMs, **RL for LLM Post-training**, **LLM Agents**, **LLM Evaluation**

SELECTED PUBLICATIONS

1. Your Benchmark Is an Empirical Measure Over Difficulty
Yifan Sun†, Naicheng Yu†, Jiawen Gong†, Jingyan Shen, Huan Zhang
NeurIPS 2026.

2. When Reasoning Meets Its Laws
Junyu Zhang†, **Yifan Sun**†, Tianang Leng†, Jingyan Shen†, Ziyin Liu, Paul Pu Liang, Huan Zhang
NeurIPS 2026.
Efficient Reasoning Workshop at NeurIPS 2025 (Oral Presentation; Best Paper Nomination).

3. SAFTAC: Simulation-Augmented Fine-Tuning of Open-Source LLMs for Analog Circuit Design
Junsheng Huang†, **Yifan Sun**†, Zhuoer Zhang†, Ning Wei, Yening Liu, Michael Molter, Pavan Kumar Hanumolu, Elyse Rosenbaum, Bin Hu, Huan Zhang
NeurIPS 2026.

4. Understanding Reasoning from Pretraining to Post-Training
Jingyan Shen†, Ang Li†, Salman Rahman, **Yifan Sun**, Micah Goldblum, Matus Telgarsky, Pavel Izmailov
NeurIPS 2026.

5. MonitorBench: A Comprehensive Benchmark for Chain-of-Thought Monitorability in Large Language Models
Han Wang†, **Yifan Sun**†, Brian Ko†, Mann Talati, Jiawen Gong, Zimeng Li, Naicheng Yu, Xucheng Yu, Wei Shen, Vedant Jolly, Huan Zhang
COLM 2026.

6. Learning a Zeroth-Order Optimizer for Fine-Tuning LLMs
Kairun Zhang†, Haoyu Li†, Yanjun Zhao†, **Yifan Sun**, Huan Zhang
ICML 2026.

7. Influence-Preserving Proxies for Gradient-Based Data Selection in LLM Fine-Tuning
Sirui Chen†, Yunzhe Qi†, Mengting Ai, **Yifan Sun**, Ruizhong Qiu, Jiaru Zou, Jingrui He
ICLR 2026.

8. Improving Data Efficiency for LLM Reinforcement Fine-Tuning Through Difficulty-targeted Online Data Selection and Rollout Replay
Yifan Sun†, Jingyan Shen†, Yibin Wang†, Tianyu Chen, Zhendong Wang, Mingyuan Zhou, Huan Zhang
NeurIPS 2025.

9. MiCRo: Mixture Modeling and Context-aware Routing for Personalized Preference Learning
Jingyan Shen†, Jiarui Yao†, Rui Yang†, **Yifan Sun**, Feng Luo, Rui Pan, Tong Zhang, Han Zhao
EMNLP 2025 (Main).
Outstanding Paper Award (7 selected out of 8,174 submissions).

10. AutoScale: Scale-Aware Data Mixing for Pre-Training LLMs
Feiyang Kang†, **Yifan Sun**†, Bingbing Wen, Si Chen, Dawn Song, Rafid Mahmood, Ruoxi Jia
COLM 2025.

11. The Emperor's New Clothes in Benchmarking? A Rigorous Examination of Mitigation Strategies for LLM Benchmark Data Contamination
Yifan Sun†, Han Wang†, Dongbai Li†, Gang Wang, Huan Zhang
ICML 2025.

12. 2D-OOB: Attributing Data Contribution through Joint Valuation Framework
Yifan Sun†, Jingyan Shen†, Yongchan Kwon
NeurIPS 2024.

13. Get More for Less: Principled Data Selection for Warming Up Fine-Tuning in LLMs
Feiyang Kang, Hoang Anh Just†, **Yifan Sun**†, Himanshu Jahagirdar†, Yuanzhi Zhang, Rongxing Du, Anit Kumar Sahu, Ruoxi Jia
ICLR 2024.

14. Data Acquisition: A New Frontier in Data-centric AI
Lingjiao Chen, Bilge Acun, Newsha Ardalani, **Yifan Sun**, Feiyang Kang, Hanrui Lyu, Yongchan Kwon, Ruoxi Jia, Carole-Jean Wu, Matei Zaharia, James Zou
Journal of Data-centric Machine Learning Research (DMLR).

SELECTED PREPRINTS AND WORKSHOP PAPERS

1. No Hidden Prompts Needed! You Can Game AI Peer Review with Presentation-Only Revisions
Xu Yang, Zhizhou Sha, Junbo Li, Jian Yu, **Yifan Sun**, Matthew Zhao, Jinrui Fang, Xinyue Guo, Yining Wu, Xu Hu, Yifu Luo, Qiang Liu, Zhangyang Wang
Preprint.

2. Direct Preference Optimization with Rating Information: Practical Algorithms and Provable Gains
Luca Viano, Ruida Zhou, **Yifan Sun**, Mahdi Namazifar, Volkan Cevher, Shoham Sabach, Mohammad Ghavamzadeh
Preprint.

3. SVIP: Towards Verifiable Inference of Open-source Large Language Models
Yifan Sun†, Yuhang Li†, Yue Zhang, Yuchen Jin, Huan Zhang
Socially Responsible and Trustworthy Foundation Models Workshop at NeurIPS 2025 (Oral Presentation).

INDUSTRY EXPERIENCE

Meta *Menlo Park, CA*
Research Scientist Intern, FAIR *May 2026 – Aug 2026*

· Identified and characterized the substantially greater robustness of distillation pre-training to overfitting relative to standard cross-entropy pre-training under repeated data, and developed empirical scaling laws to inform data reuse and compute allocation.

Amazon.com, Inc. *Sunnyvale, CA*
Applied Scientist Intern, AGI Foundations *May 2025 – Aug 2025*

· Proposed SPO, a generalized preference optimization objective using adaptive skewed reference distributions and margins, provided theoretical insights that unify existing RLHF algorithms, and demonstrated superior empirical performance over 5 strong baselines.

SELECTED RESEARCH EXPERIENCE

ASTRAL Lab, UIUC
Advisor: Prof. Huan Zhang *Aug 2024 – Present*

· Developed a theoretically grounded difficulty-aware framework to diagnose when LLM benchmarks may produce misleading model rankings or miss localized training progress; analyzed 25 benchmarks across 7 domains and introduced BENCHFILL to reduce the sum of the three largest difficulty-coverage gaps by 21.09%–38.88% with only 3 added items per benchmark.

· Presented the Laws of Reasoning (LoRe), a unified framework that formalizes intrinsic reasoning patterns in large reasoning models by studying reasoning compute and accuracy as two fundamental axes and operationalizing them through monotonicity and compositionality properties.

· Proposed Difficulty-targeted Online daTa Selection (DOTS) and Rollout Replay (RR) to improve data efficiency in LLM reinforcement fine-tuning, achieving a 23%–62% reduction in fine-tuning time across 6 LLM-dataset combinations while matching standard GRPO performance.

· Developed a rigorous evaluation pipeline with two novel metrics, fidelity and contamination resistance, to systematically assess 20 mitigation strategies for LLM benchmark data contamination across 5 benchmarks and 10 LLMs.

ReDS Lab, Virginia Tech
Advisor: Prof. Ruoxi Jia *Aug 2023 – May 2024*

· Developed AutoScale, a scaling law-based domain reweighting technique to optimize data composition for pre-training LLMs at target data scales, reaching a target validation perplexity at least 28% faster than baseline methods and improving downstream task performance across 8 benchmarks.

· Introduced GOT-D, a scalable data selection method based on Optimal Transport for efficient LLM pre-fine-tuning, reducing model toxicity by over 0.2 with only 20K samples and enhancing downstream performance on domain-specific and zero-shot tasks.

SKILLS & ACADEMIC SERVICE

Languages	English (Proficient), Chinese (Native)
Coding	Python, R, C, C++, SQL
Tools	PyTorch, TensorFlow, Transformers, verl, vLLM, Accelerate
LLM Expertise	Pre-training, Supervised Fine-tuning, RLHF, RLVR, Inference, Optimization
Reviewer	NeurIPS, ICML, ICLR, TMLR, EMNLP, COLM, AAAI